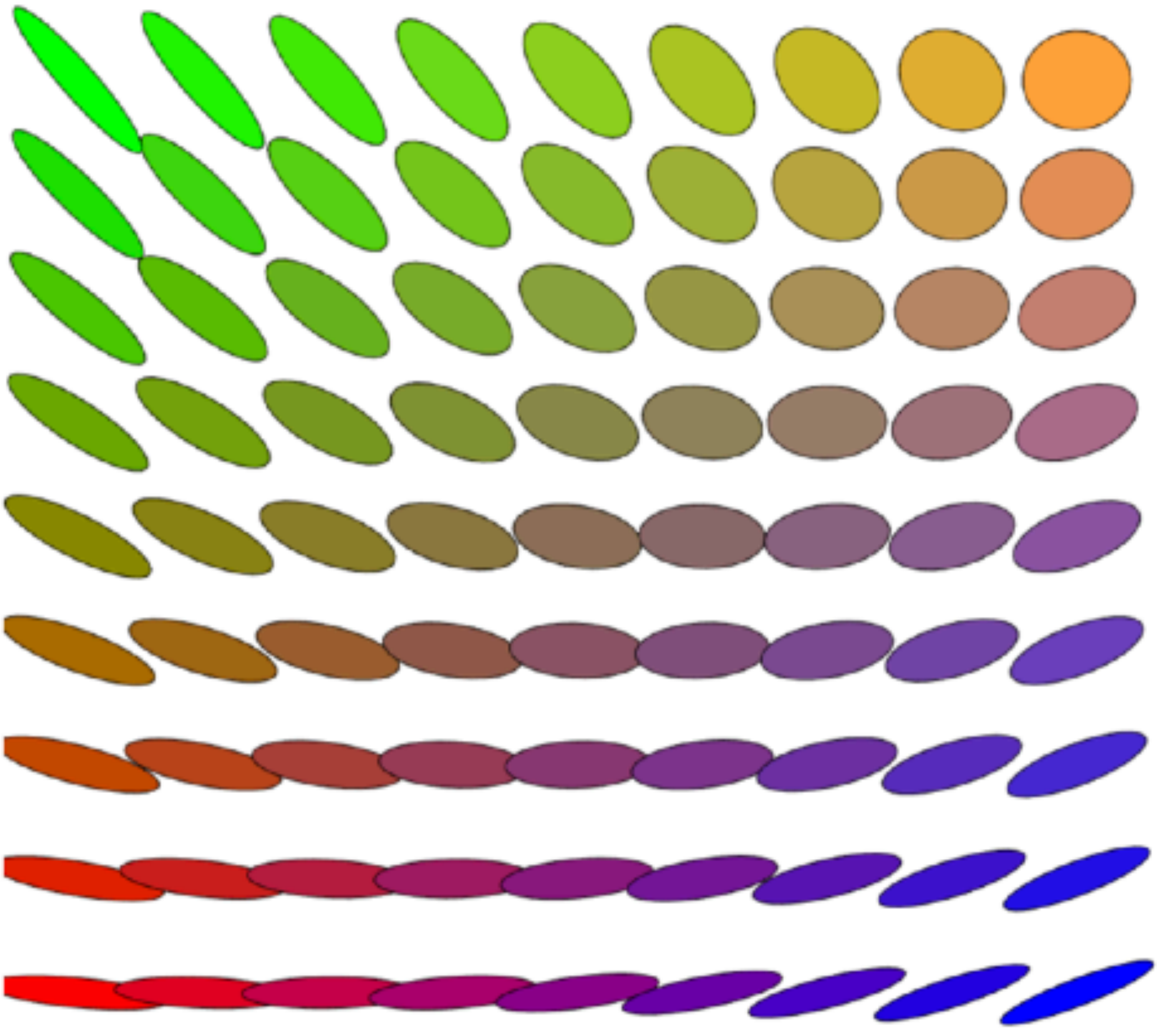
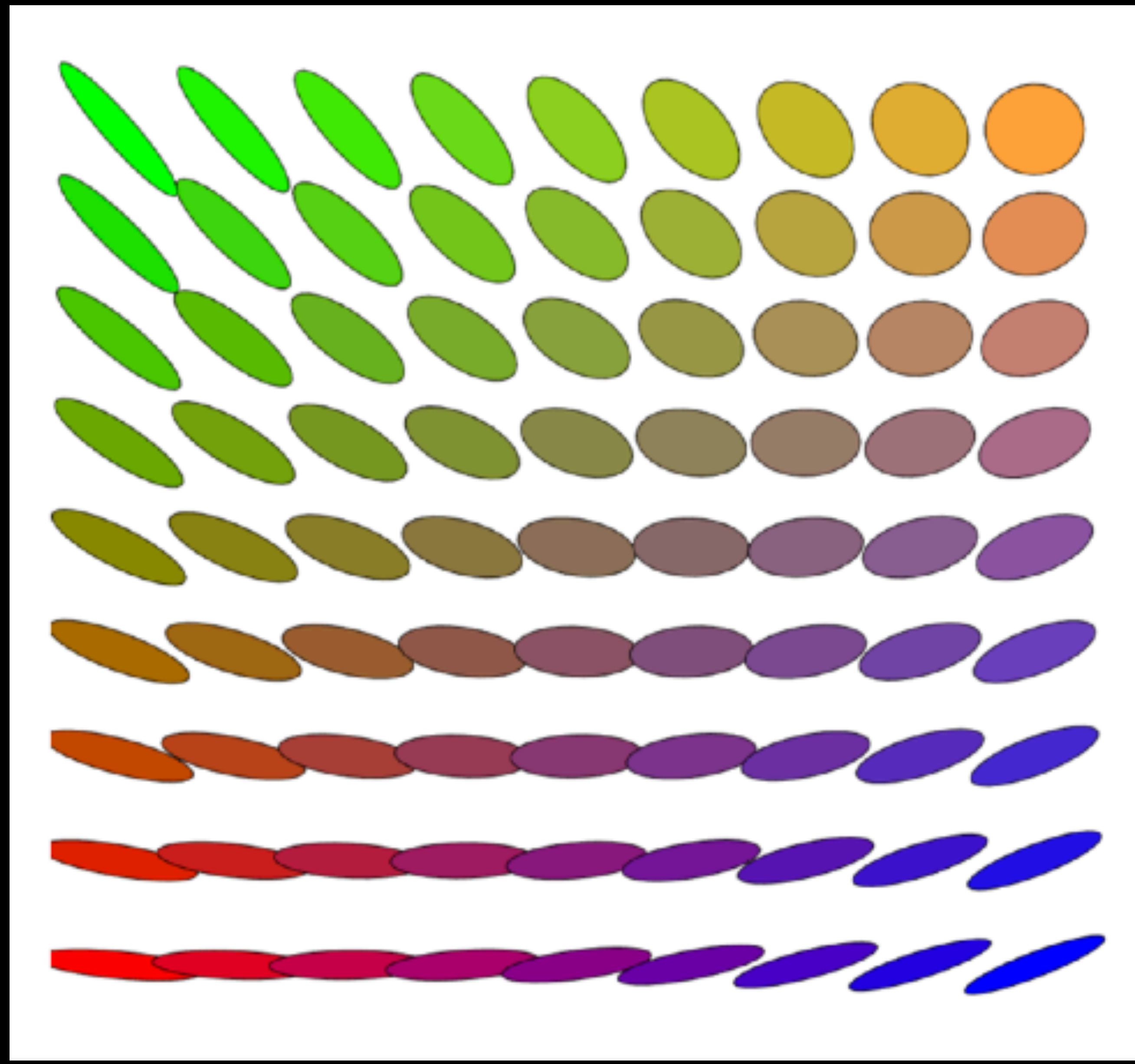


Example: Gaussians



- Suppose $\mu_i = \mathcal{N}(m_i, \Sigma_i)$
- Then barycenter $\nu^\star = \mathcal{N}(m^\star, \Sigma^\star)$ with:
  - Mean: $m^\star = \sum_i \lambda_i m_i$ (Euclidean mean).
  - Covariance: $\Sigma^\star$solves matrix equation (nonlinear):
    $\Sigma^\star = \arg\min_{\Sigma} \sum_i \lambda_i \mathscr{B}(\Sigma_i, \Sigma)^2$ for $\mathscr{B}$ the Bures metric
- Demonstrates geometry is nontrivial in variance/covariance.

Geometric Intuition